

# Mini-project 1: Global Temperature

Autumn 2026 · Team project · Report 3–5 pages + runnable code + AI log

## Overview

How fast is the world warming, how certain are we, and what does the warming co-vary with? In this project your team downloads a real, open global surface temperature record, quantifies the warming trend and its uncertainty, and tests how the temperature series correlates with greenhouse-gas concentrations and world population. Along the way you will confront the difference between correlation and causation deliberately, with at least one absurd variable. Work as a team! Figure out together how to read and plot the data and discuss together what a good representation is. The same dataset can be represented in many ways. Choose a visualization that makes the scientific question clear without hiding variability or uncertainty. **Think carefully about what your figure should help the reader see.** Be creative in presentation but make sure that the representation is scientifically faithful to the data.

## Learning goals

- obtain and document an observational climate dataset
- visualize updated state of the art temperature anomalies together with observational uncertainty
- estimate a warming trend and quantify how the estimate depends on the analysis period
- distinguish measurement uncertainty from variability and regression uncertainty
- combine time series with different temporal resolution
- explain why statistical correlation does not by itself establish physical causation
- use physical mechanisms and independent evidence when evaluating causal explanations
- produce a reproducible analysis in which data, code and AI assistance are documented.

## Data sources

The primary dataset is the **Berkeley Earth global land + ocean temperature record** (Rohde & Hausfather, 2020):

[https://berkeley-earth-temperature.s3.us-west-1.amazonaws.com/Global/Land\\_and\\_Ocean\\_complete.txt](https://berkeley-earth-temperature.s3.us-west-1.amazonaws.com/Global/Land_and_Ocean_complete.txt)

The file gives monthly global mean temperature anomalies in °C relative to the 1951–1980 average, from 1850 to the present, with uncertainties quoted as 95% confidence intervals. Header lines start with the comment character %. The file contains two versions that differ in how sea ice is treated: one uses air temperature above sea ice, the other uses water temperature below sea ice. Use the “air temperature above sea ice” version as the standard (it is the first series in the file, and the variant normally quoted for global warming). If you prefer, HadCRUT5 or a similar record may be used instead — document your choice.

Supporting datasets for the correlation questions:

**CO<sub>2</sub> concentration** (atmospheric, monthly and annual global means, ppm) — NOAA Global Monitoring Laboratory: [https://gml.noaa.gov/ccgg/trends/gl\\_data.html](https://gml.noaa.gov/ccgg/trends/gl_data.html) (e.g. the files `co2_mm_g1.csv` and `co2_annmean_g1.csv`).

**CH<sub>4</sub> concentration** (monthly and annual global means, ppb) — NOAA GML: [https://gml.noaa.gov/ccgg/trends\\_ch4/](https://gml.noaa.gov/ccgg/trends_ch4/)

**CO<sub>2</sub> emissions** (annual, global and by country) — Our World in Data's CO<sub>2</sub> and Greenhouse Gas Emissions database: <https://ourworldindata.org/co2-emissions> (complete dataset as CSV via their GitHub repository, [github.com/owid/co2-data](https://github.com/owid/co2-data)). Note that emissions (Gt CO<sub>2</sub>/yr) and concentration (ppm) are different quantities — be clear about which you use.

**World population** (annual) — UN World Population Prospects:

<https://www.un.org/development/desa/pd/content/world-population-prospects-2024-dataset>. A convenient ready-made CSV of the same UN series is available from Our World in Data:

<https://ourworldindata.org/grapher/population-unwpp>

**For every dataset you use, record the URL, the date you downloaded it, and the dataset version.** Cite the data in your report and include the code that reads it.

## Part 1 — Data acquisition and first plot

- Download and read the Berkeley Earth land/ocean record with your own code (Do not editing of the file).
- Plot the monthly anomaly with its uncertainty as a shaded band. Label the band correctly: the file quotes 95% confidence intervals.
- Add a 12-month moving average.
- Describe the dataset and give a good visual representation with readable axes, units, and a caption that lets the figure stand on its own.



*"Warming stripes": each stripe is one year's global mean temperature anomaly, 1850 to present (blue = colder, red = warmer than the 1951–1980 baseline). Concept: Ed Hawkins, [showyourstripes.info](https://showyourstripes.info). One example of creative presentation. Can you find your own?*

## Part 2 — Driving questions

### Q1. What is the linear rate of warming?

Fit a linear trend to the temperature record and report it in °C/decade, with uncertainty. You must choose a fit window: Use 1980–2025 as your primary trend period. Then repeat the analysis using start years 1970 and 1990. How much does the estimated warming rate depend on the chosen interval, and why?

Adjacent monthly temperatures are not statistically independent. A warm month tends to be followed by another relatively warm month because the climate system has memory. Ordinary least-squares regression therefore tends to underestimate the uncertainty if each monthly value is treated as an independent observation. For this project, calculate annual means before fitting the trend.

### Q2. Is the warming linear — or accelerating?

Has the observed warming rate changed? Compare the linear trend for 1970–1997 with that for 1998–2025 or use two other periods justified in advance. Discuss whether differences necessarily imply a change in the underlying forced warming rate.

Fit a quadratic (or fit linear trends to an early and a late window) and report the acceleration with its uncertainty, again stating the start and end years. What do the sign and size of the acceleration mean physically?

Fit a quadratic as a simple empirical test for curvature. Do not assume that global temperature is physically expected to follow a quadratic function.

### **Q3. What co-varies with the warming?**

Correlate the temperature series with atmospheric CO<sub>2</sub>, with CH<sub>4</sub>, and with world population. The series have different time resolutions and coverage. Then harmonize them (e.g. annual means over the common period) before computing any correlation coefficient. Report the correlations, and discuss: which of these relationships could be causal, in which direction, and through what physical mechanism? What confounding factors link them to each other?

### **Q4. The chocolate test**

Now correlate the temperature series with something that certainly does not drive the climate — for example world chocolate consumption, or another steadily growing quantity of your choice. You will very likely find an impressively high correlation. Use this result to discuss why a correlation coefficient alone can never establish causation, and what additional physical reasoning (e.g. radiative forcing by greenhouse gases) separates Q3 from Q4.

## **Interpretation, robustness and reporting**

**Short report (3–5 pages):** data and sources, methods, main findings with uncertainties, limitations, and the code included or linked.

**AI log:** what was done with AI, which prompts were used, which suggestions were accepted or rejected, and why.

## **Assessment criteria (suggested)**

**Reproducibility & code** Clean, commented code and a runnable notebook or script. Does the code run for the others on your team?

**Data documentation.** Clear sources (URL/DOI, download date), version, license/metadata. Can your teammates find the same data?

**Analysis** Correct handling of time resolution (month vs year), trends with windows and units, uncertainties, correlations and possible lags.

**Interpretation,** well-reasoned conclusions, an honest discussion of causality vs correlation, and possible confounding factors.

**Presentation** — figures with axes, units and captions; a short, clear report.

## **Common pitfalls**

Mixing monthly and annual series without harmonizing them first. Correlations require harmonized datasets.

Overinterpreting correlation, especially population vs temperature (that is what Q4 is for).

Quoting a trend without specifying the period and the unit (°C/decade).

Mislabelling the uncertainty band (the Berkeley Earth file gives 95% confidence intervals, not  $\pm 1\sigma$ ).

Confusing CO<sub>2</sub> emissions (Gt CO<sub>2</sub> per year) with CO<sub>2</sub> concentration (ppm).

Missing data references, URLs, or dataset versions.

## Ethical use of AI (project requirement)

AI may be used for generating code, figures, and information search, provided that:

- What tool has been used
- You explain clearly how you have used AI tools,
- All code is attached and runnable,
- It is clear which parts the
- work that AI contributed to
- You understand and can explain your results.

## References

Rohde, R. A., & Hausfather, Z. (2020). The Berkeley Earth Land/Ocean Temperature Record. *Earth System Science Data*, 12, 3469–3479. <https://doi.org/10.5194/essd-12-3469-2020>

Lan, X., Tans, P., & Thoning, K. W.: Trends in globally-averaged CO<sub>2</sub>. NOAA Global Monitoring Laboratory. <https://gml.noaa.gov/ccgg/trends/>

Lan, X., Thoning, K. W., & Dlugokencky, E. J.: Trends in globally-averaged CH<sub>4</sub>. NOAA Global Monitoring Laboratory. [https://gml.noaa.gov/ccgg/trends\\_ch4/](https://gml.noaa.gov/ccgg/trends_ch4/)

United Nations, Department of Economic and Social Affairs, Population Division (2024). *World Population Prospects 2024*. <https://population.un.org/wpp/>

Hawkins, E. Warming stripes. <https://showyourstripes.info>